INCH-POUND

MIL-DTL-24231/4E(SH)
w/AMENDMENT 1
13 May 2024
SUPERSEDING
MIL-DTL-24231/4E(SH)
26 April 2013

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, TYPE I, MOLDED, 14-, 24-, 30-, OR 40-CONDUCTOR
(STRAIGHT AND 90 DEGREES)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 11.

2. For cable wiring connections, see applicable table in appendix A of MIL-DTL-24231.

3. Sleeve M24231/3-010 may be modified prior to assembly by drilling or slotting to allow cable entry from the side. Sharp edges shall be rounded.

4. Ni-Cu plug sleeves shall be coated with plasma spray non-conductive material in accordance with S9320-AM-PRO-030/MLDG.

5. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

Distribution Statement A. Approved for public release: distribution is unlimited.

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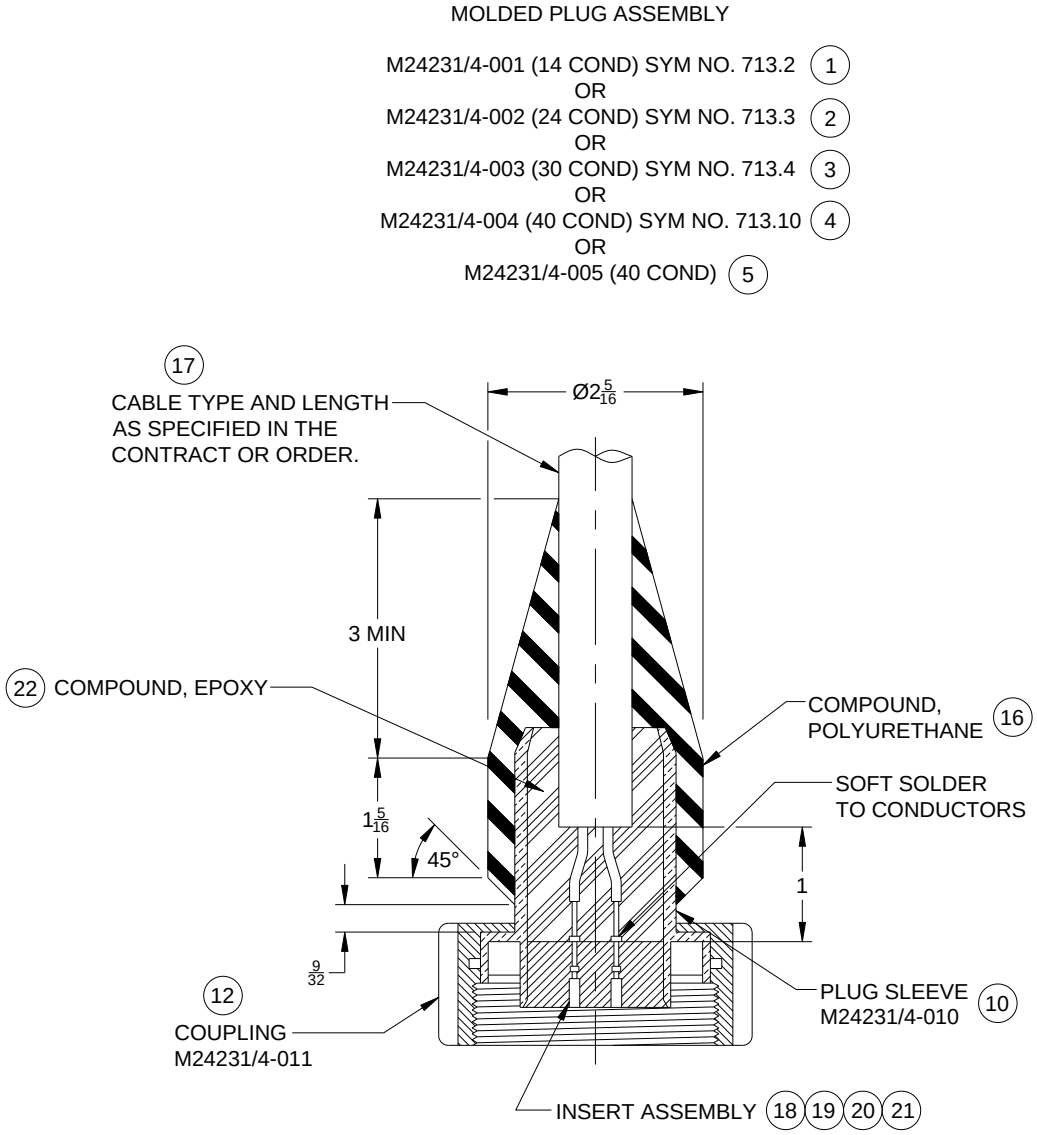


FIGURE 1. Connector, plug, molded – typical.

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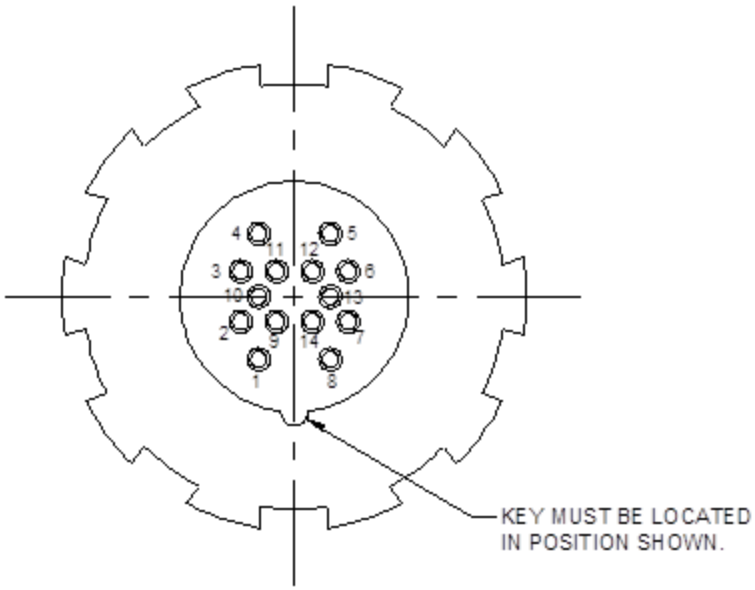
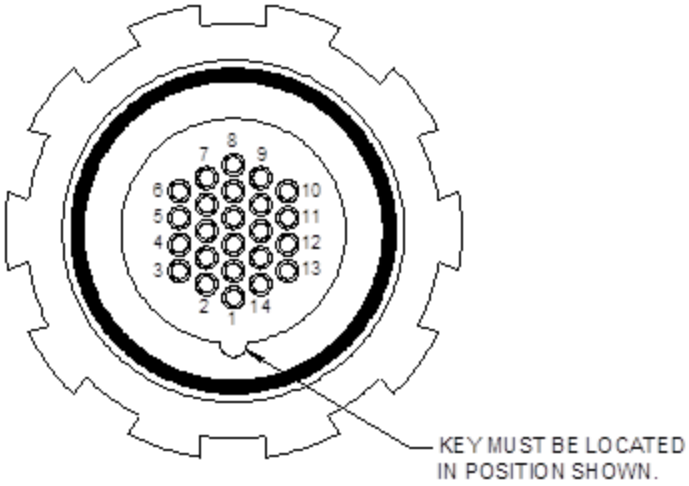


FIGURE 2. Connector, plug, 14-conductor (straight) M24231/4-001. (1)

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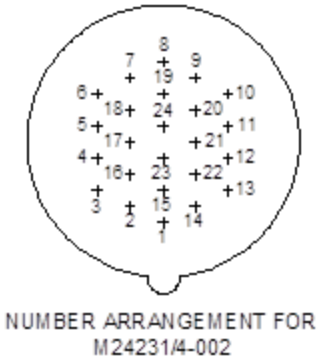
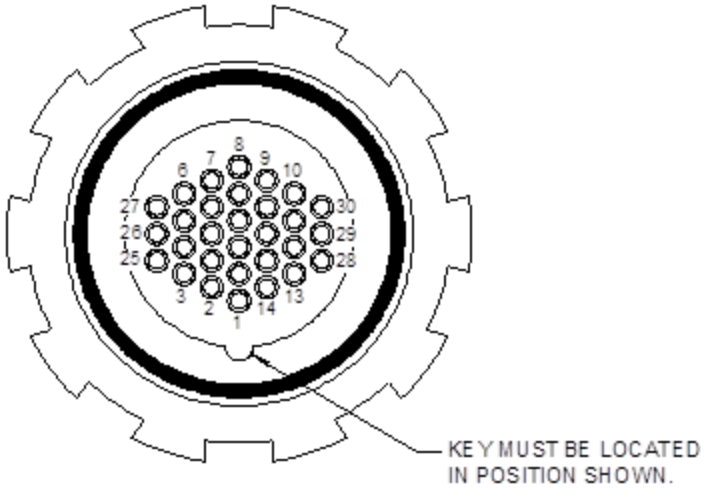


FIGURE 3. Connector, plug, molded, 24-conductor (straight) M24231/4-002. (2)

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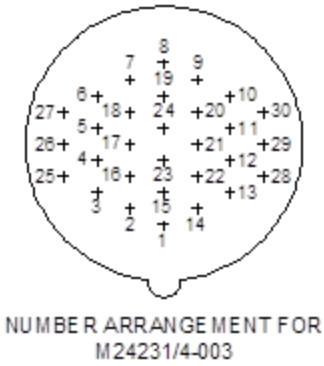


FIGURE 4. Connector, plug, molded, 30-conductor (straight) M24231/4-003. (3)

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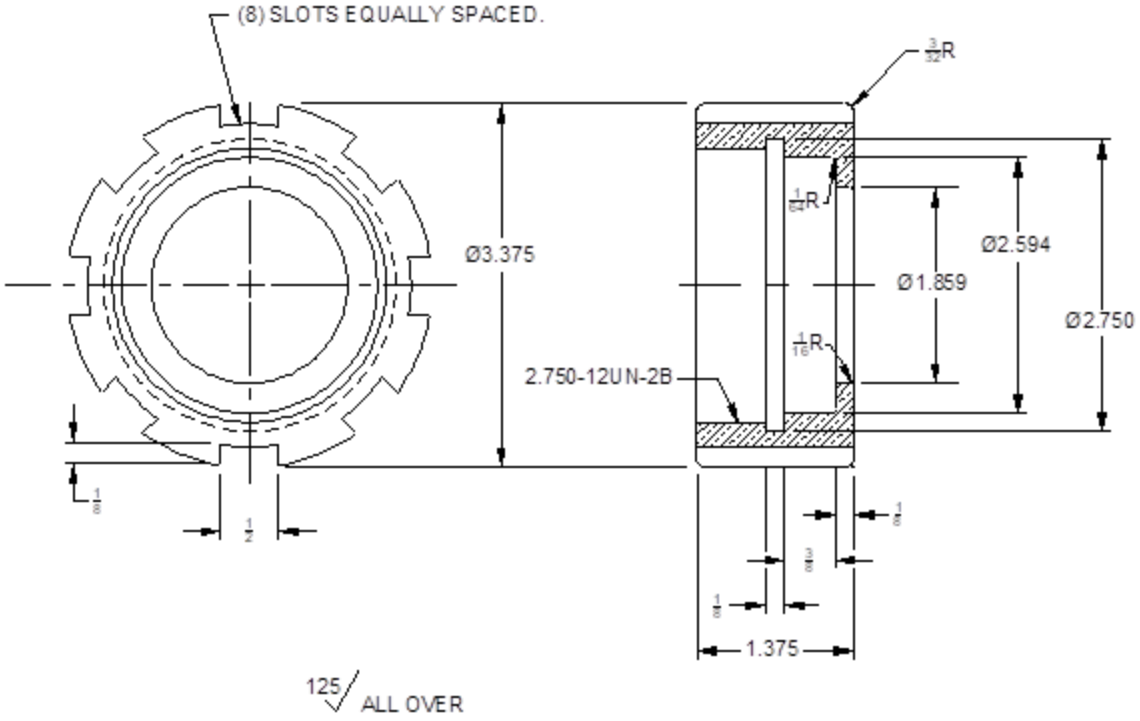


FIGURE 5. Coupling – M24231/4-011. (12)

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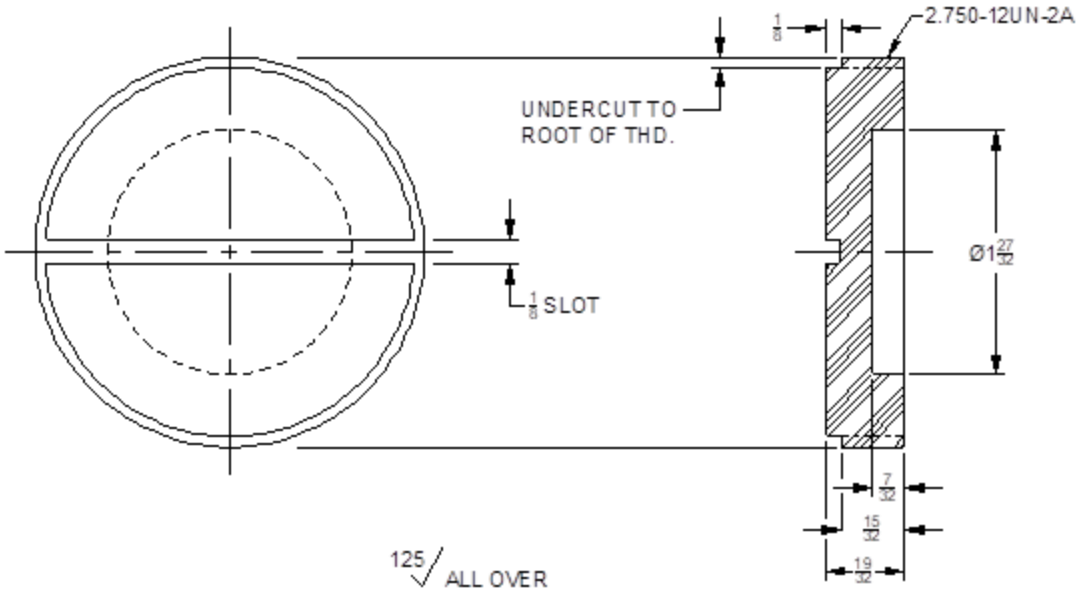


FIGURE 6. Protective cap – M24231/4-012. (11)

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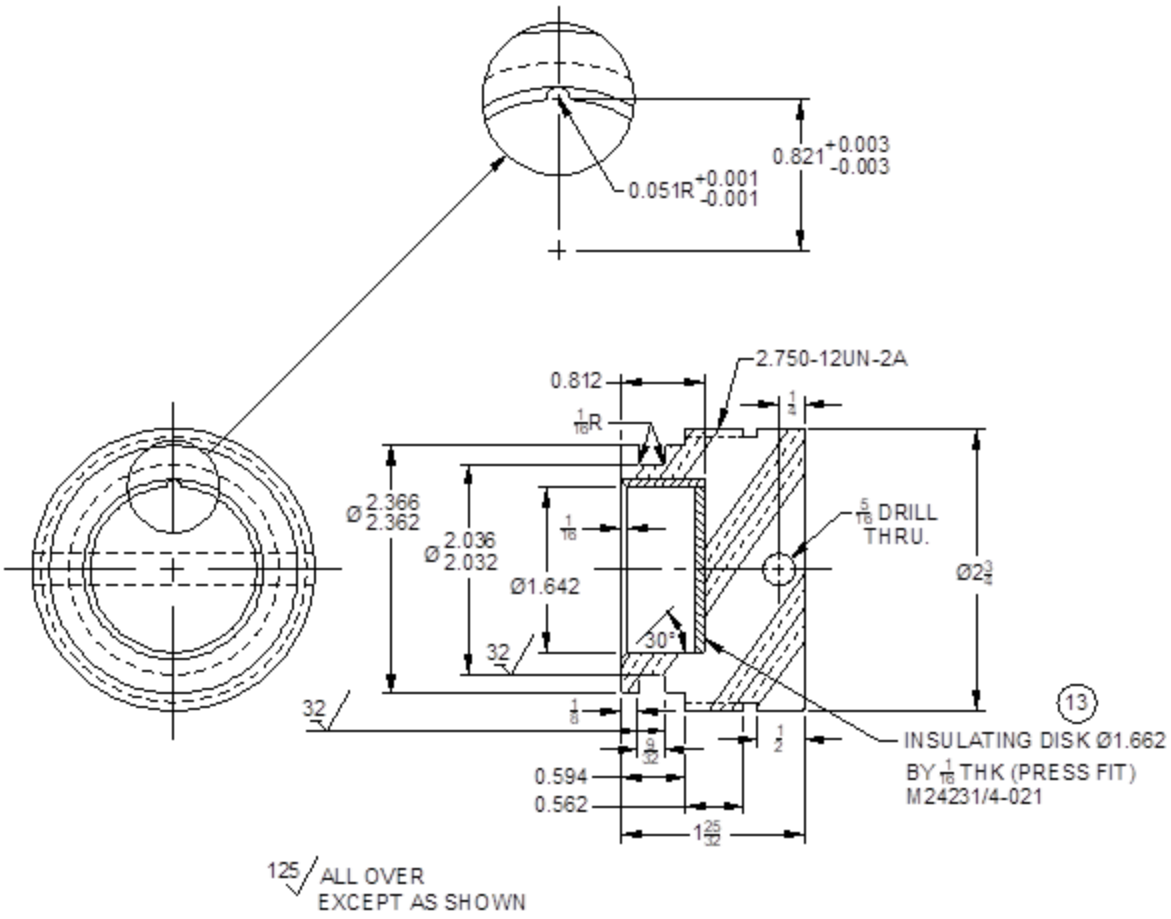


FIGURE 7. Sealing plug. (15)

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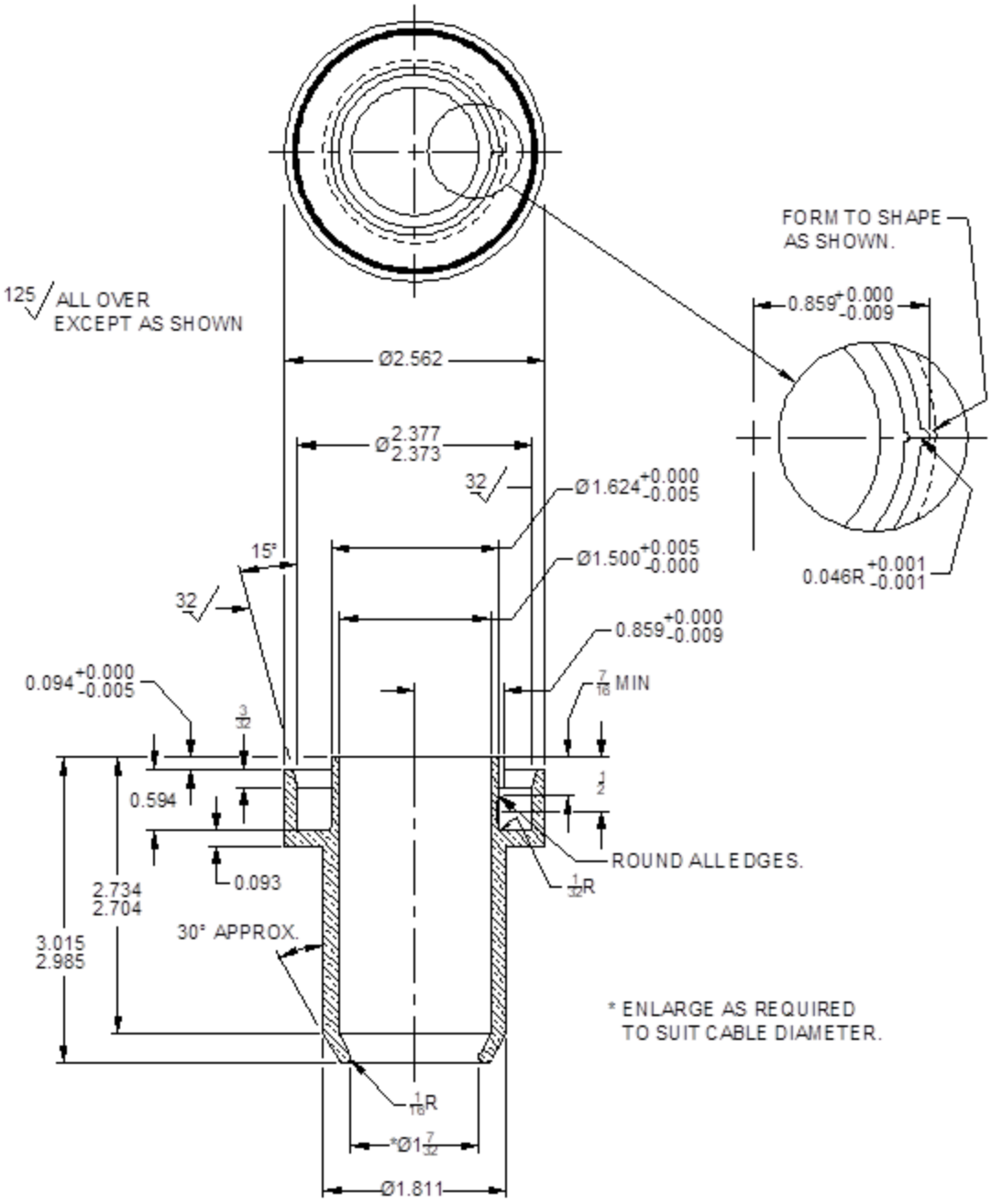


FIGURE 8. Plug sleeve – M24231/4-010. (10)

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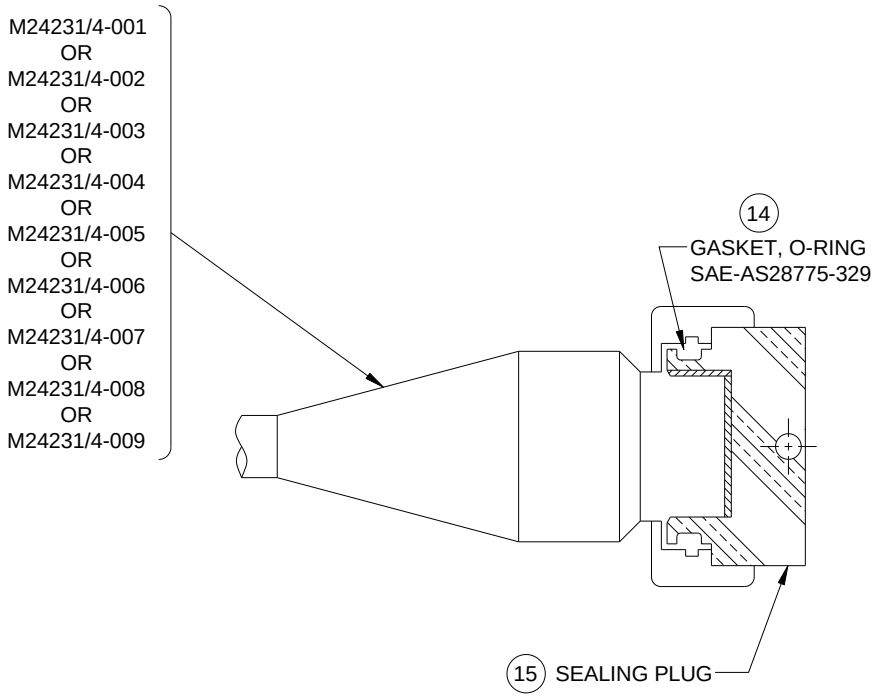
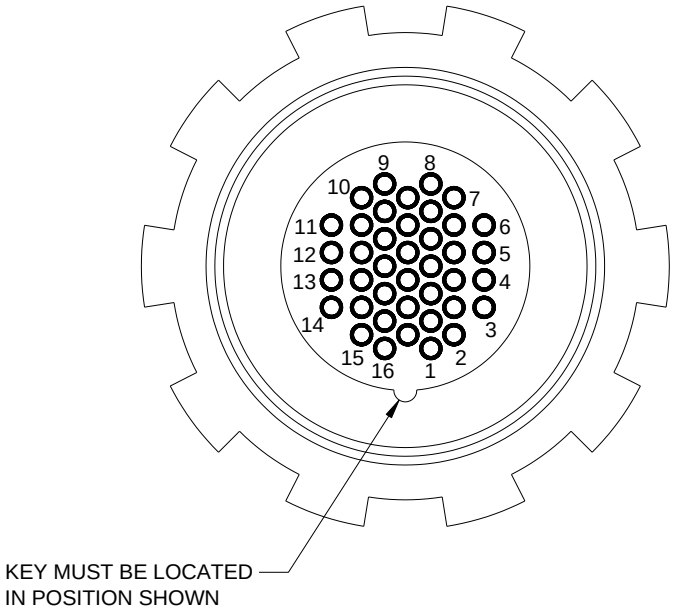
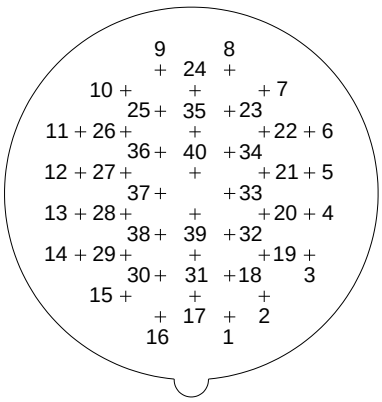


FIGURE 9. Sealing plug assembly – M24231/4-020. 23

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NOTE: M24231/4-004 AND M24231/4-005 ARE PHYSICALLY THE SAME. THE PART NUMBERS DEFINE USAGE, (SEE APPENDIX A OF MIL-DTL-24231).

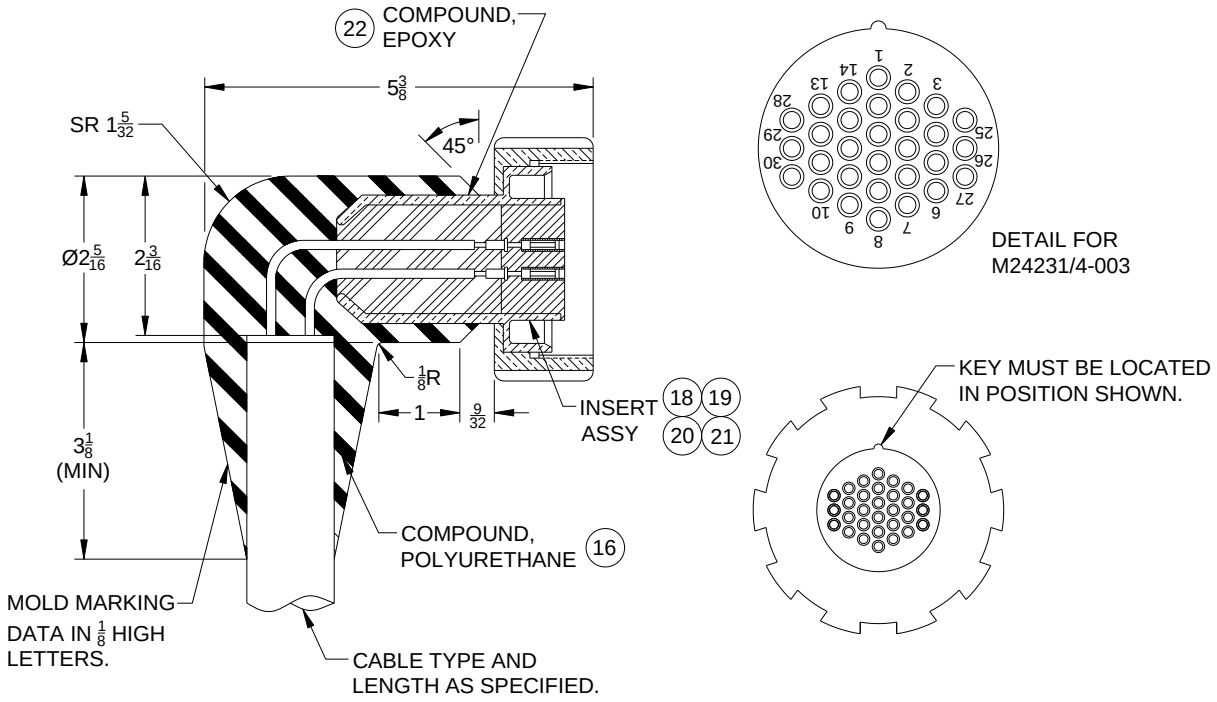




NUMBER ARRANGEMENT FOR
M24231/4-004 OR M24231/4-005

FIGURE 10. Connector, plug, molded, 40-conductor – M24231/4-004, M24231/4-005, and M24231/4-009. (4)

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14 CONDUCTOR (6)
90° PLUG ASSEMBLY
M24231/4-006
SAME AS M24231/4-001
EXCEPT AS SHOWN.

24 CONDUCTOR (7)
90° PLUG ASSEMBLY
M24231/4-007
SAME AS M24231/4-002
EXCEPT AS SHOWN.

30 CONDUCTOR (8)
90° PLUG ASSEMBLY
M24231/4-008
SAME AS M24231/4-003
EXCEPT AS SHOWN.

40 CONDUCTOR (9)
90° PLUG ASSEMBLY
M24231/4-009
SAME AS M24231/4-004
EXCEPT AS SHOWN.

FIGURE 11. Connector, plug, molded, 14-, 24-, 30-, and 40-conductor (90-degree).

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TABLE I. Parts list.

Item no. [1]	Description [2]	Quantity for M24231/4- [3]									Material/specification	Remarks
		001	002	003	004	005	006	007	008	009		
1	Plug assembly, 14-conductor, symbol no. 713.2	1									[4]	See figure 2.
2	Plug assembly, 24-conductor, symbol no. 713.3		1								[4]	See figure 3.
3	Plug assembly, 30-conductor, symbol no. 713.4			1							[4]	See figure 4.
4	Plug assembly, 40-conductor, symbol no. 713.10				1						[4]	See figure 10.
5	Plug assembly, 40-conductor					1					[4]	See note on figure 10.
6	Plug assembly, 90 degrees, 14-conductor						1				[4]	See figure 11.
7	Plug assembly, 90 degrees, 24-conductor							1			[4]	See figure 11.
8	Plug assembly, 90 degrees, 30-conductor								1		[4]	See figure 11.
9	Plug assembly, 90 degrees, 40-conductor									1	[4]	See figure 11.
10	Sleeve, plug	1	1	1	1	1	1	1	1	1	Ni-Cu, class A, QQ-N-281	See figure 8 and requirement 3.
11	Cap, protective	1	1	1	1	1	1	1	1	1	Plastic, type MAI-60, ASTM D5948 or polycarbonate, ASTM D3935 [6]	Nonpressure-proof, see figure 6.
12	Coupling	1	1	1	1	1	1	1	1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 5.
13	Disk, insulating, 1.662 diameter by 1/16 thick	1	1	1	1	1	1	1	1	1	Non-laminated plastic, composition A, type II, grade GU, L-P-535	See figure 7.

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TABLE I. Parts list – Continued.

Item no. [1/]	Description [2/]	Quantity for M24231/4- [3/]									Material/specification	Remarks
		001	002	003	004	005	006	007	008	009		
14	Gasket, O-ring	1	1	1	1	1	1	1	1	1	SAE-AS28775-329	Lubricate with silicone compound, SAE-AS8660; see figure 9.
15	Sealing plug	1	1	1	1	1	1	1	1	1	Ni-Cu, class A, QQ-N-281	Pressure-proof, see figure 3 and figure 7.
16	Compound, polyurethane	As required									[7/]	See figure 1 and figure 11.
17	Cable	As required									[8/]	See figure 1.
18	Insert assembly, 14-conductor	1					1				M24231/29-006	See figure 1.
19	Insert assembly, 24-conductor		1					1			M24231/29-007	See figure 1.
20	Insert assembly, 30-conductor			1					1		M24231/29-008	See figure 1.
21	Insert assembly, 40-conductor				1	1				1	M24231/29-009	See figure 1.
22	Compound, epoxy	As required									Cast epoxy resin compound [9/]	Base EE4183 or RE2039, hardener HD-3561; see figure 1 and figure 11.
23	Sealing plug assembly [10/]	1	1	1	1	1	1	1	1	1	See items 14 and 15	See figure 9.

NOTES:

[1/] Item numbers are the same as encircled numbers shown on figures.

[2/] See applicable figure or table listed in remarks column for part numbers and dimensions.

[3/] Quantity is for one complete assembly of each size.

[4/] When plugs are purchased as a kit, the inserts are not molded in accordance with MIL-DTL-24231/29. The socket contacts and sleeving are individually provided.

[5/] DELETED.

[6/] May be Ni-Al-BRZ in accordance with ASTM B150/B150M at the option of the manufacturer.

[7/] Molding compound to be procured in accordance with S9320-AM-PRO-020/MLDG.

[8/] For cable type, see table for cross index of mating plugs, receptacles, adapters, and single and multiple cable connectors of MIL-DTL-24231.

[9/] Hysol base EE4183 or RE2039, hardener HD-3561, Hysol Division, Dexter Corporation, Olean, NY, CAGE 04347, or equal.

[10/] To be supplied when specified (see acquisition requirements of MIL-DTL-24231).

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BACKWARDS COMPATIBILITY. Connectors with glass reinforced epoxy (GRE) plug sleeves procured to MIL-DTL-24231/4E, dated 26 April 2013, are not acceptable for shipboard use and should be disposed of. All future requisitions should be to the current revision of this specification sheet and MIL-DTL-24231.

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5935-2024-112)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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